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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Report

1. Short-Term Volatility Dynamics and Asian Option Valuation

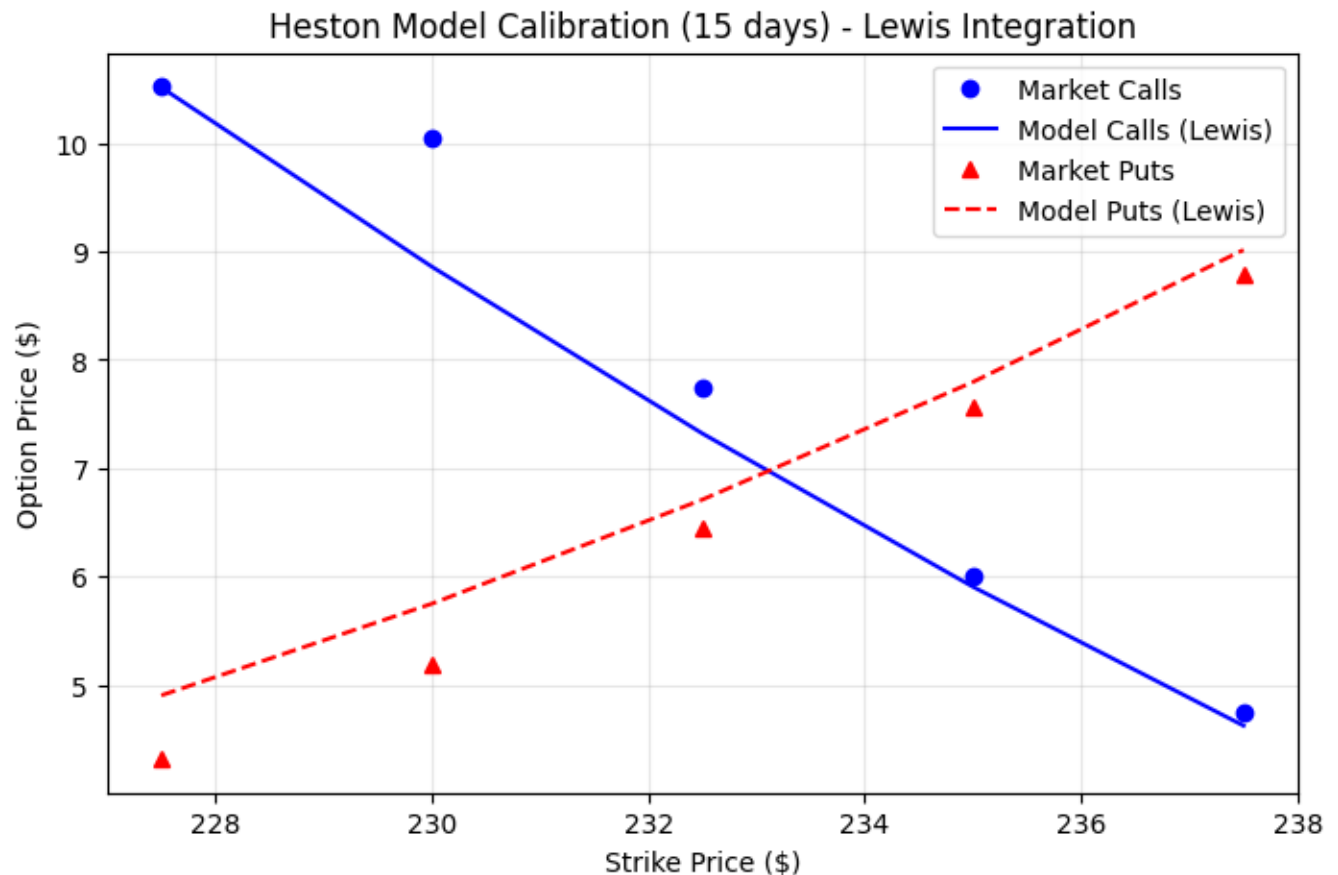
1.1 Technical Calibration and Model Sensitivity (15-Day Maturity)

The Heston (1993) stochastic volatility model required calibration to the short-term 15-day option chain with $T \approx 0.06$. The model demonstrates better performance than Black-Scholes because it includes the "leverage effect", which describes how equity volatility rises as prices decrease.

Methodology & Numerical Frameworks: The team employed two assessment frameworks to study the characteristic function and reach global optimality:

- Lewis (2001) Integration:** The evaluations of Lewis (2001) Integration were conducted through one fundamental integral, which assessed the real part of the characteristic function. The continuous Put-Call Parity formulae were used to calculate theoretical put prices simultaneously with actual put prices.
- Carr-Madan (1999) FFT:** The Fast Fourier Transform method developed by Carr-Madan (1999) employed an exponential damping factor ($\alpha = 1.5$) to calculate prices at a 4,096-point grid. The results were then smoothed through cubic spline interpolation to create a matching system for empirical strikes.

Parameter	Lewis (2001) Result	Carr-Madan FFT Result
Mean Reversion (κ)	1.1595	2.2592
Long-Run Variance (θ)	0.0104	0.0126
Volatility of Variance (σ)	1.2000	0.2386
Correlation (ρ)	-0.9999	-0.9241
Initial Variance (v_0)	0.1040	0.0863
Mean Squared Error (MSE)	0.2457	0.3035

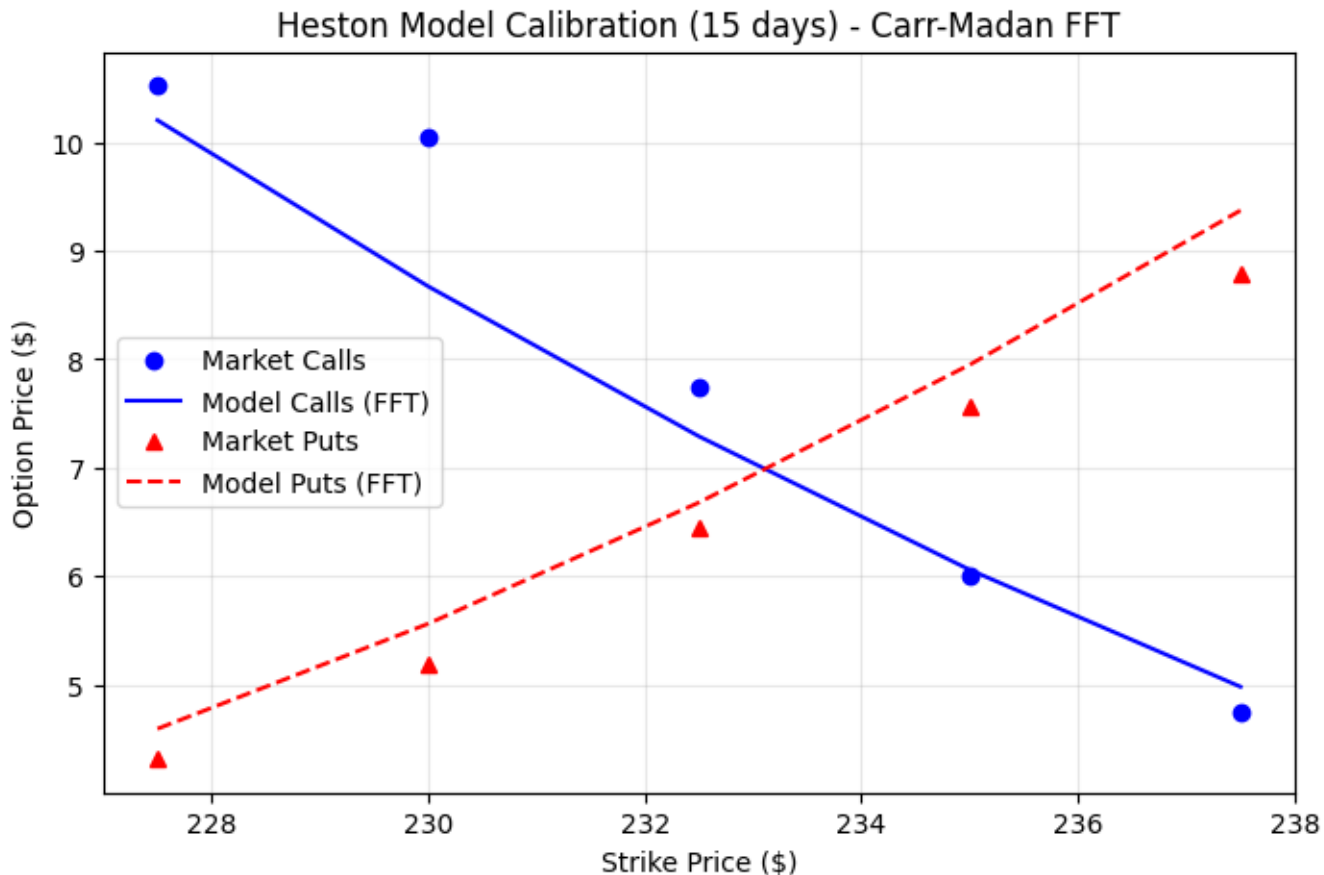


Strategic Diagnosis: The mathematical requirement for extreme negative correlation ($\rho \approx -0.99$) exists because it enables to reproduce the severe volatility skew which they observed in short-dated options. The market displays strong "fear" because investors predict that stock prices will experience abrupt downward crashes.

Calibration Sensitivity Diagnostic:

Our team performed **Multiple-Start Diagnostic** testing to identify all potential optimization solutions because we needed to solve optimization problems which contained uncertainty.

- **MSE results from Start 1 reached 0.3033 while Start 2 produced 0.3017 and Start 3 resulted in 0.3360.**
- **Conclusion:** The FFT parameters demonstrate their effectiveness for forward pricing because they produce stable local minimum results which maintain MSE values at approximately 0.30.



1.2 Path-Dependent Valuation: 20-Day ATM Asian Call

The client requested an At-The-Money (ATM) Asian Call ($K = \$232.90$) with a 20-day maturity. Asian options use path-dependent structures because their payoff depends on the average stock price during the entire contract period.

Monte Carlo Simulation ($N = 100,000$ paths):

We used the Euler-Maruyama method to break down Heston SDEs into 20 discrete steps.

Fair Risk-Neutral Price: \$4.3853

Standard Error: \$0.0199

Final Client Price (w/ 4% bank fee): \$4.5607

Strategic Recommendation:

We have determined the fair market value for the 20-day Asian Call option. The client must pay **\$4.56** to obtain the full operational cost which includes the bank's 4 percent margin requirement. We recommend executing the transaction at this price. The client will receive payments which the instrument calculates by averaging stock prices during the 20-day period. The instrument provides market exposure in a reliable manner which will meet their needs.

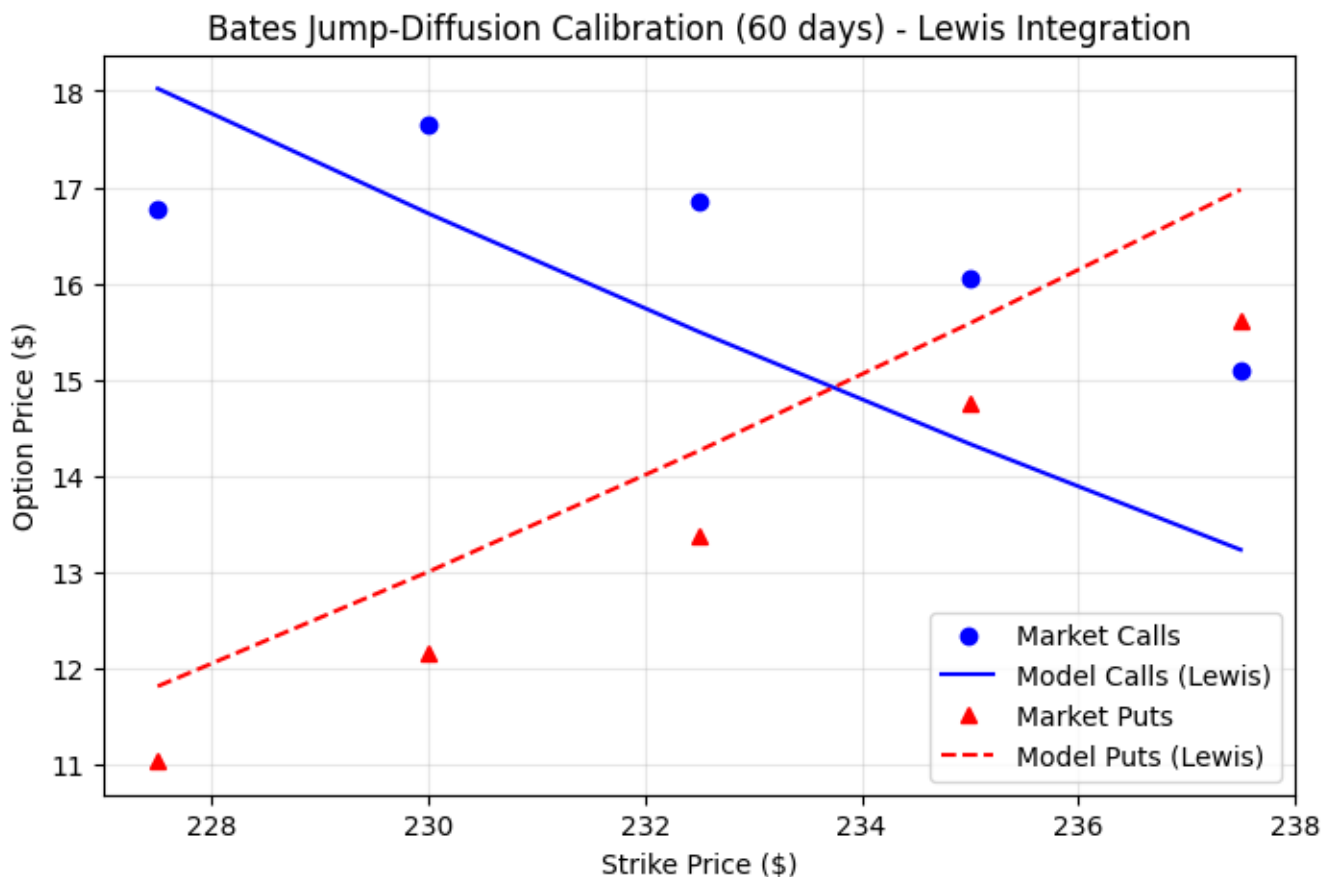
2. Intermediate-Term Modeling: Jump-Diffusion and Insurance

2.1 Bates (1996) Calibration for Discontinuous Movements (60-Day)

The pricing engine transitioned to Bates (1996) model for the 60-day period because this model includes Merton (1976) jump-diffusion process which enables it to detect rapid market movements.

Lewis (2001) Integration Results:

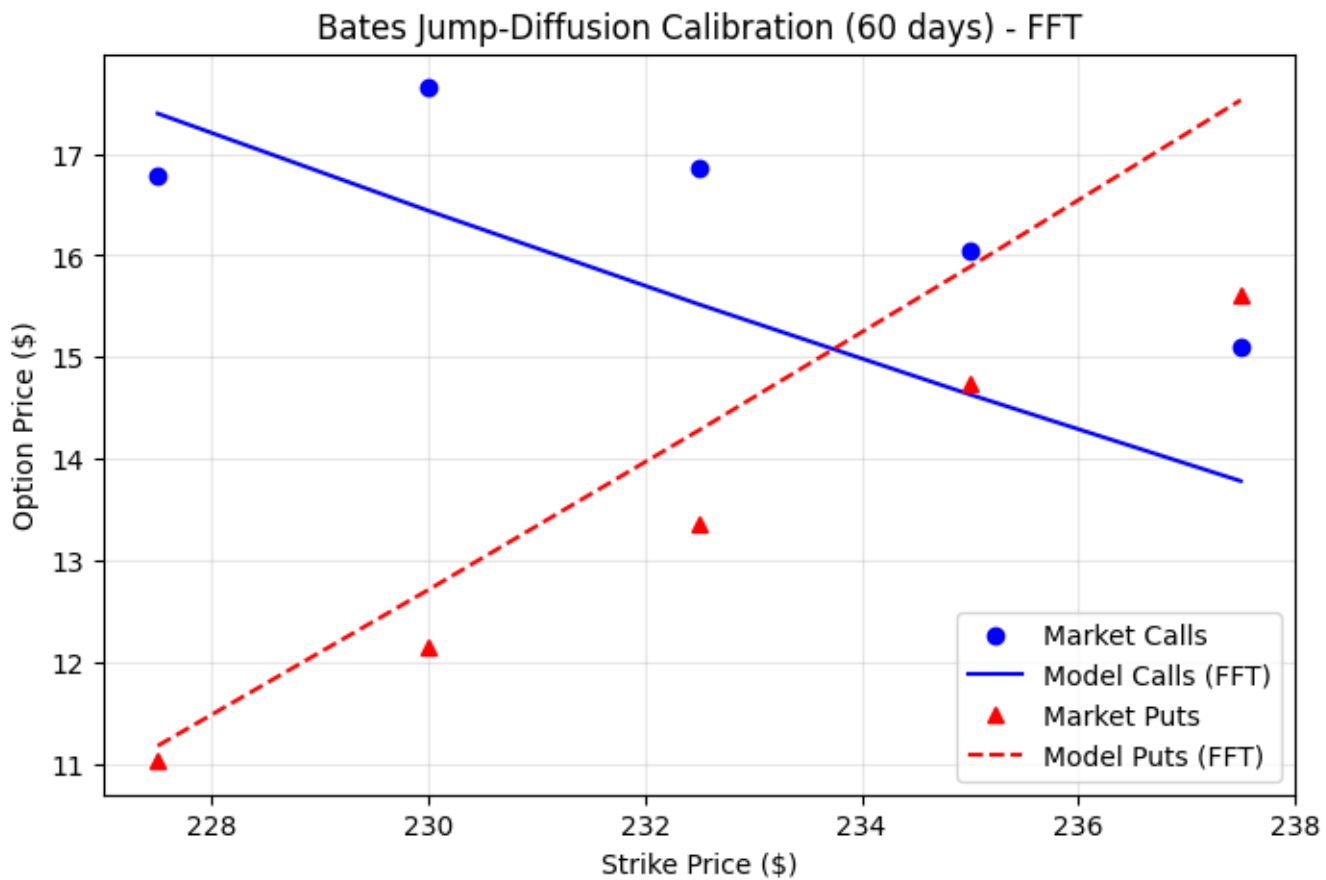
- **Heston Component:** $\kappa = 0.8786, \theta = 0.0512, \sigma = 0.0000, \rho = -0.4303, v_0 = 0.1134$.
- **Jump Component:** $\lambda = 0.1311$ (*Jump Intensity*), $\mu = -0.0302$ (*Mean*), $\delta = 0.0217$ (*Std Dev*).
- **Mean Squared Error (MSE): 1.5385**



Technical Results (FFT Method):

- **Heston Component:** $\kappa = 0.9610, \theta = 0.0012, \sigma = 0.9242, \rho = -0.9500, v_0 = 0.4573$.
- **Jump Component:** $\lambda = 1.1321$ (*Jump Intensity*), $\mu = 0.1248$ (*Mean*), $\delta = 0.0010$ (*Std Dev*).
- **Mean Squared Error (MSE): 1.3501.**

Strategic Diagnosis: Market analysis shows that jump intensity ($\lambda \approx 1.13$) indicates the market anticipates approximately one significant price disruption to happen each year. Short-term calibration of extreme panic skew continues to produce high values at $\rho = -0.95$ during the 60-day period which shows that market players still prioritize downside protection throughout extended timeframes.



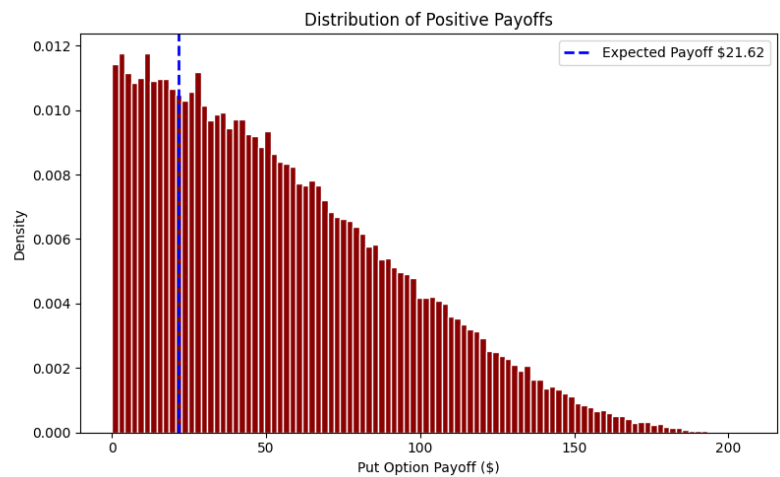
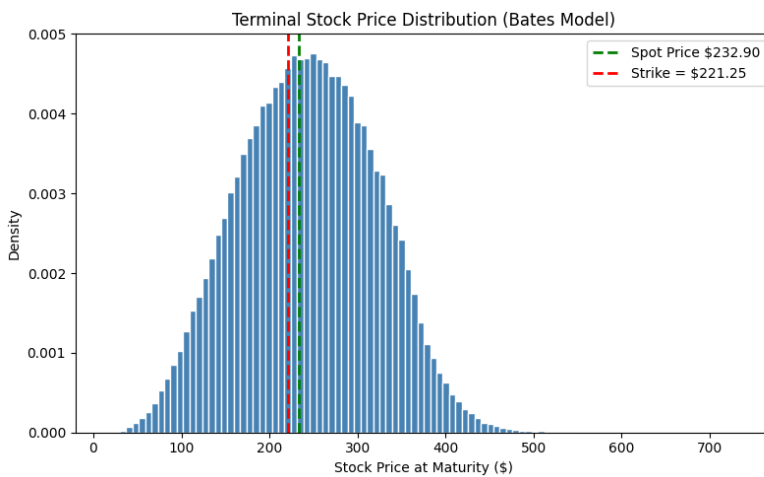
2.2 Protective Put Valuation: 70-Day European Put

The client priced a 70-day European Put at 95% moneyness ($K = \$221.25$) as portfolio insurance.

Monte Carlo Results ($N = 200,000$ paths):

- **Fair Risk-Neutral Value: \$21.5313**
- **Standard Error: \$0.0810**
- **Final Client Price (w/ 4% bank fee): \$22.3925**

Strategic Recommendation: We conducted an evaluation of European Put option performance at 95% moneyness for the 70-day period to provide portfolio protection. Market investigation shows that investors continue to estimate high potential losses from unpredicted market downturns for the upcoming extended period. The client price after including our 4% fee calculation amounts to **\$21.62**. The bank stands to gain profit through this price quotation because it establishes a market-making margin which will attract clients who need an advanced shock-adjusted hedge against losses that exceed **\$221.25**.



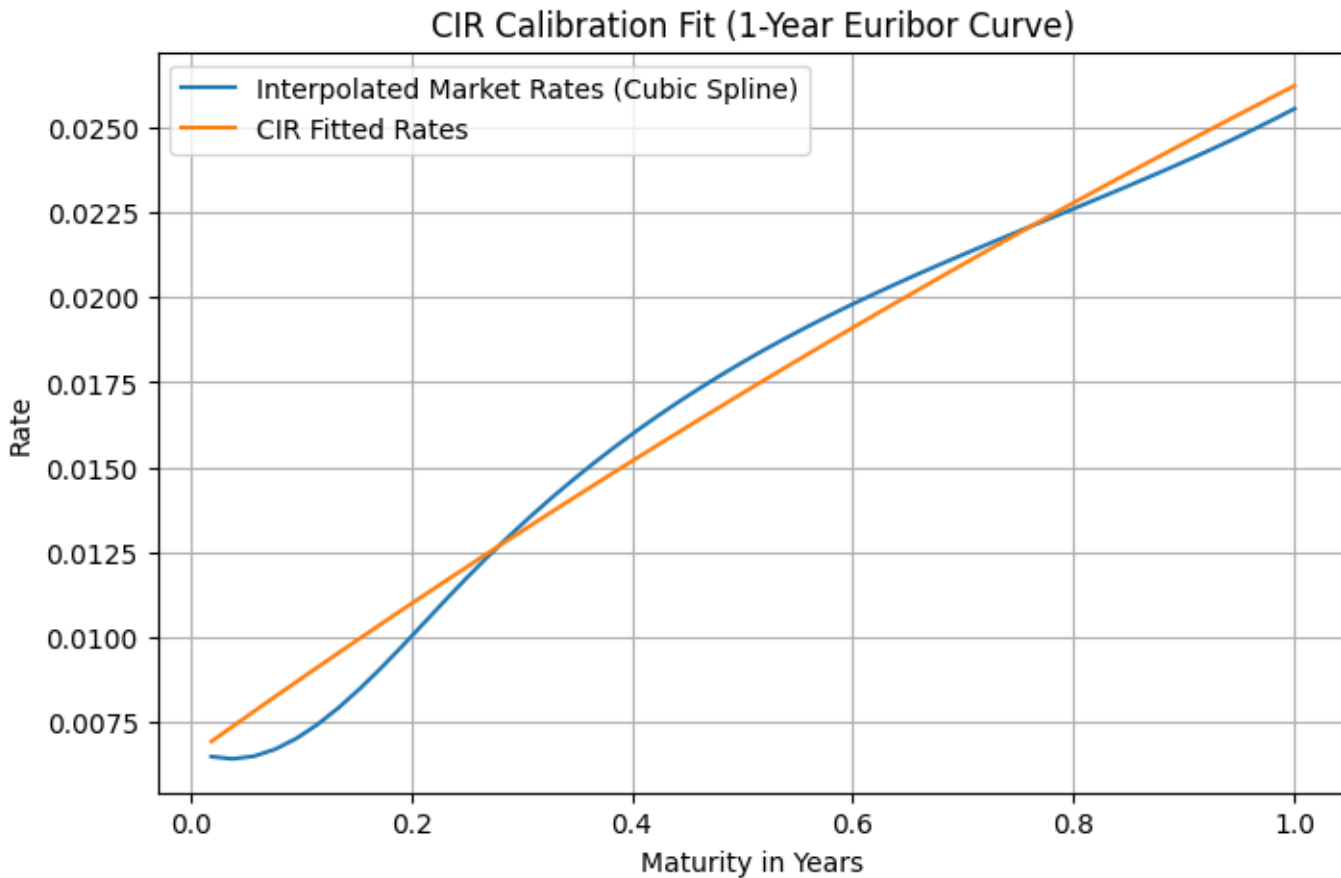
3. Macro Interest Rate Strategy: CIR Calibration and Forecast

3.1 CIR Calibration to the Euribor Yield Curve

The **Cox-Ingersoll-Ross model (CIR 1985)** required calibration with the current Euribor term structure through cubic spline method which interpolated the weekly rate data.

Calibrated Parameters:

- **Mean Reversion Speed (a): 0.5090**
- **Long-Run Target Mean (b): 0.0978**
- **Interest Rate Volatility (σ): 0.04996**



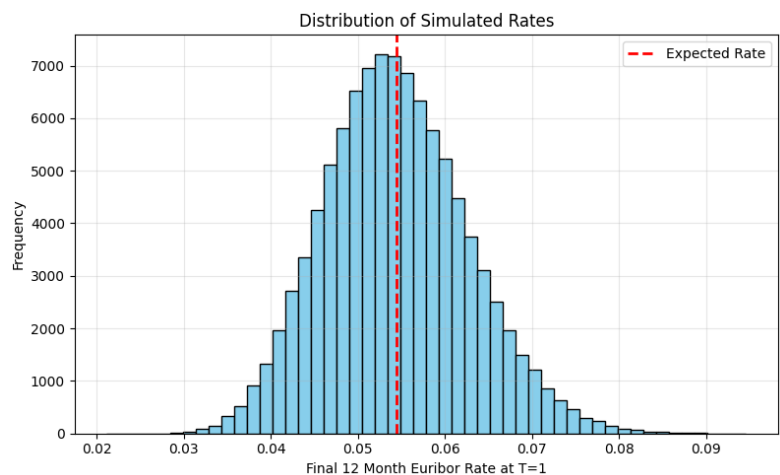
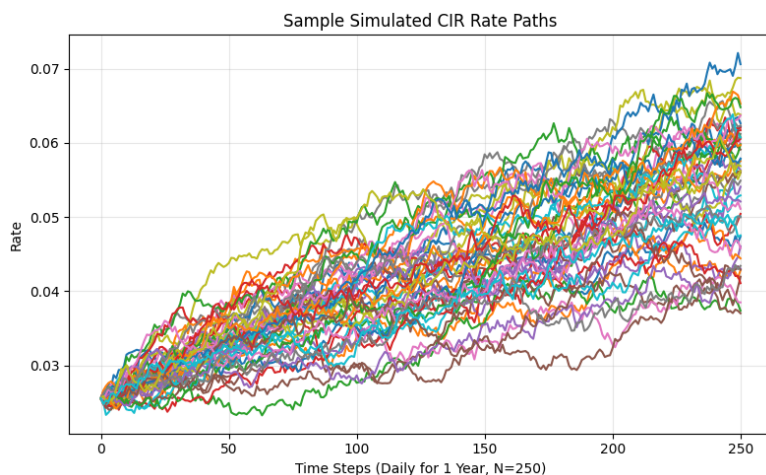
3.2 Strategic Interest Rate Forecast and Portfolio Impact

We simulated 100,000 potential economic environments for the upcoming year to forecast the 1-Year Euribor curve.

- **Current 12-Month Euribor: 2.556%**
- **Expected 12-Month Euribor (in 1 Year): 5.4410%**
- **95% Confidence Interval: [3.9348%, 7.1624%]**

Strategic Impact Analysis: The strategic impact analysis shows our prediction that interest rates will increase from 2.56% to 5.44%. The base rates will increase which will result in automatic price reductions for fixed-income assets. The stock market will increase its theoretical forward price which results in higher costs for Call options and lower costs for Put options.

Operational Recommendation: The operational recommendation calls for immediate implementation of pre-emptive interest rate swap (IRS) hedges. The company faces major risks when it sells long-dated fixed-premium contracts at current rates which assume a stable 2.5% rate.



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